

Question 1 Final Report — Strategic Lending Forecasting & LLM Benchmarking

Executive Summary

The Question 1 deliverables provide a full-stack forecasting and reporting environment for Strategic Lending. The deterministic balance-sheet engine enforces accounting identities while a TensorFlow forecaster adds ratio-based drivers, macro features, and probabilistic heads for scenario generation. The Part 2 LLM workflow now benchmarks prompt datasets, evaluates coverage and errors, and synthesizes CFO-ready recommendations that default to deterministic forecasts when LLM responses lack quantitative output. Recent benchmarking runs introduce structured JSON output enforcement and higher token budgets, lifting numeric coverage for `gpt-4o-mini` to over 80% across 99 prompts. This report consolidates Part 1 and Part 2 findings, documents the chosen methods, and records testing evidence for correctness and reproducibility.

1. Literature Review Highlights

Key literature themes that shaped the solution: - **Identity-preserving projections.** Vélez-Pareja & Mejía-Pelaez informed the asset-liability-equity tie-out and financing-plug logic. - **Driver-based forecasting.** Financial-ratio and growth-driver approaches motivated the feature design (liquidity, leverage, profitability, spreads, and lagged deltas) for forward projections. - **Probabilistic scenario modelling.** Bayesian/variational forecasting and ensemble techniques informed the Gaussian and variational heads plus calibrated bank ensembles for uncertainty. - **LLM quantitative benchmarking.** Recent evaluation frameworks for structured numeric outputs informed coverage/MAE/MAPE evaluation.

The repository includes a dedicated literature summary and specification documents to preserve citation notes and design rationale.

2. Method Selection & Rationale

The project combines deterministic constraints with learned forecasting: - **Deterministic balance-sheet engine** ensures $\text{Assets} = \text{Liabilities} + \text{Equity}$ and creates a reliable backbone for underwriting use. - **Neural forecaster** adds flexibility to model nonlinear relationships between ratios, growth, and macro history, while probabilistic heads enable scenario ranges. - **Calibrated bank ensembles** incorporate bank-specific liability mix priors to stabilize outputs for financial institutions. - **LLM benchmarking** provides a transparent assessment of model coverage and error before introducing LLMs into a regulated forecasting workflow.

This hybrid strategy balances interpretability (deterministic accounting) with predictive power (ML/LLM augmentations).

3. Data Acquisition & Feature Engineering

- **Data sources.** Financial statements for nine tickers (AAPL, MSFT, GM, JPM, BAC, C, HON, CAT, UNP) are downloaded via reproducible pipelines and cached as raw JSON.
- **Feature construction.** The driver dataset blends ratio metrics (liquidity, leverage, profitability), growth signals, and lagged covariates.
- **Validation.** Dataset validation checks filing gaps, duplicates, and missing columns prior to training and evaluation.

4. Part 1 Results — Forecasting & Identity Preservation

4.1 Deterministic + ML Forecasting

The deterministic constraint layer reconciles financing gaps and preserves accounting identities across forecast horizons. The TensorFlow forecaster produces multi-target balance-sheet predictions with bank-aware routing and shared feature towers.

4.2 Forecast Accuracy (Latest Two Statement Pairs)

Mean absolute errors (billions USD) and identity gaps averaged over the latest two statement pairs:

Ticker	Mode	Assets MAE (B)	Equity MAE (B)	Net Income MAE (B)	Identity Gap (B)
AAPL	mlp	17.74	17.52	67.98	0.000000
BAC	bank_ensemble	<0.01	<0.01	32.17	<0.001
C	bank_ensemble	<0.01	<0.01	14.07	<0.001
CAT	mlp	10.60	14.53	13.20	-0.000000
GM	mlp	18.52	20.44	16.45	0.000000
HON	mlp	10.28	10.58	11.20	NaN
JPM	bank_ensemble	<0.01	<0.01	59.10	<0.001
MSFT	mlp	42.64	46.48	61.47	0.000000
UNP	mlp	55.50	20.28	24.45	0.000000

Notes: bank MAEs are sub-\$10M, equity MAE is sub-\$10k, and identity gaps are below \$1M. HON's identity gap is undefined due to missing liabilities splits in the source statement.

4.3 PDF Ratio Extraction

PDF ratio extraction workflows now support multiple issuer layouts (GM, LVMH, Tencent, Alibaba, JPM, Exxon, Microsoft, VW, Google), including text-fallback parsing when table extraction fails. Provenance metadata logs source PDFs, page indexes, and table strategies to support downstream CFO narratives.

5. Part 2 Results — LLM Benchmarking & Reporting

- **Prompt datasets** pair structured statements with forecast targets.
- **LLM evaluation** computes numeric coverage, MAE, and MAPE; baseline `t5-small` produces zero numeric coverage for AAPL/BAC prompts, reinforcing the need for stronger models before reliance.
- **Comparison tooling** aligns deterministic forecasts with LLM outputs and generates recommendation memos for executive review.

Latest LLM benchmark summary (99 prompts, 3 seeds):

Model	Max New Tokens	Coverage Mean	MAE Mean (B)	MAPE Mean
gpt-4o-mini (openai-chat)	2048	81.27%	9.95	1.72

6. Testing Plan & Results

Testing follows a layered strategy: - **Unit tests** for driver features, bank template logic, and constraint enforcement. - **Integration tests** for ingestion, training, evaluation, PDF extraction, and LLM pipelines. - **Reproducibility checks** for seeded datasets and deterministic reporting outputs.

Current test coverage includes: - `pytest tests/mlcoe_q1 -q` (unit + integration for data pipelines, models, and reporting).

7. Conclusion & Next Steps

The Question 1 stack now meets the Part 1 forecasting requirements and establishes a defensible LLM benchmarking framework for Part 2. Recommended follow-ups: - Expand LLM coverage with stronger models and prompt variants. - Increase PDF ratio presets for additional issuers. - Extend probabilistic scenario support with macro-driven Monte Carlo overlays.

Appendix — Key References & Artifacts

- Deterministic specification: `reports/q1/deterministic_balance_sheet_spec.md`
- Literature summary: `reports/q1/literature_summary.md`
- Interim dashboards: `reports/q1/status/`
- LLM comparison artifacts: `reports/q1/q1_response_summary.md`